NATIONAL UNIVERSITY OF MODERN LANGUAGE ISLAMABAD

OEL report

Submitted to:

Sir Hassan Shah

Submitted by:

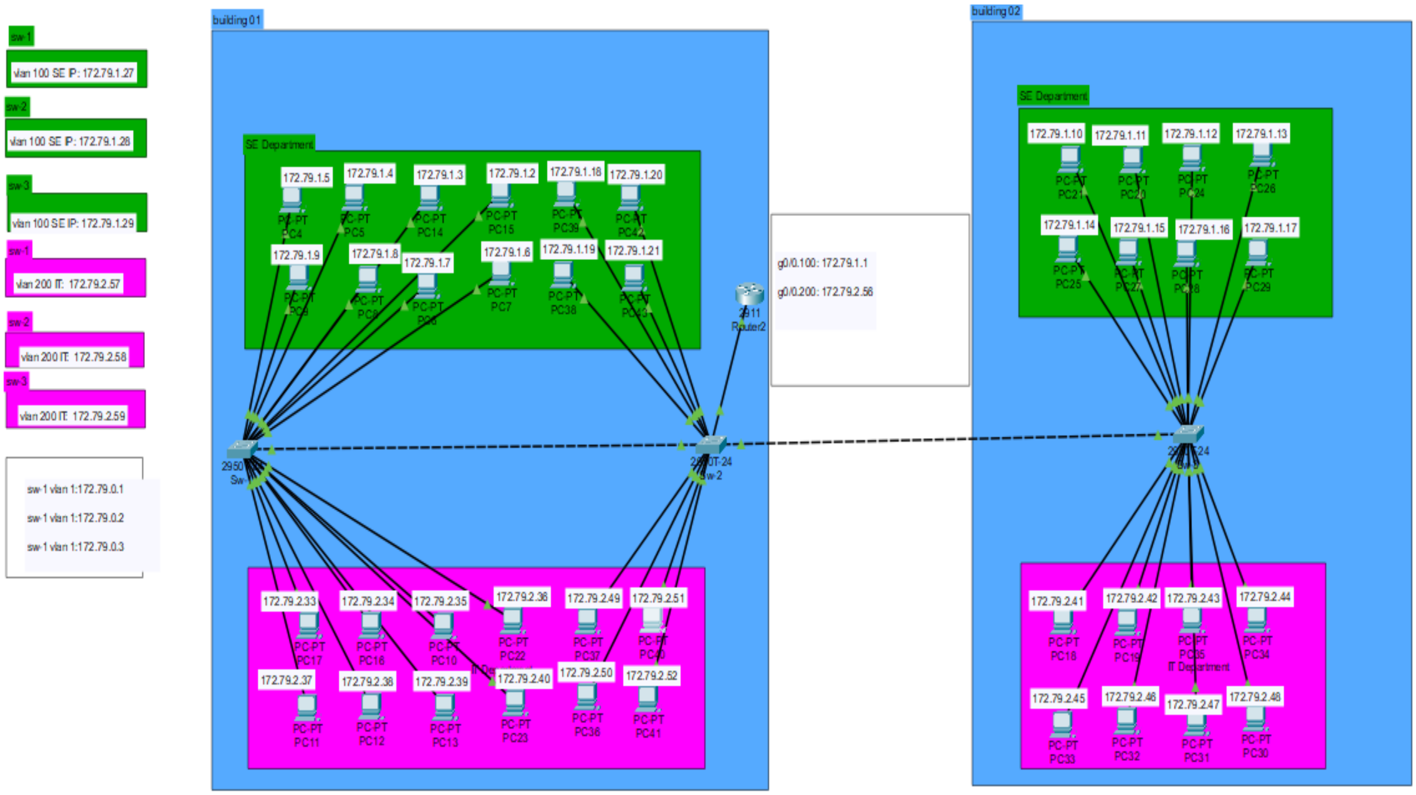
Zulqarnain Hassan

Roll No: …………………… SP21379

Subject: …………………… computer network

Department: ……………… BSSE-5-Afternoon

Date: ……………………… 03-12-2024



Step 1: Topology Design

Design the topology as shown in the figure.

- Use Cisco Packet Tracer tocreate the topology.
- Place a Router 2911 and three 2950T switches.
- Use color coding to differentiate VLANs: VLAN 100 (e.g., green) and VLAN 200 (e.g., pink).

Step 2: VLAN Creation

- Access each switch and enter global configuration mode.
- Create VLAN 100 (SE) and VLAN 200 (IT):

```
Switch>en
Switch#confi
Configuring from terminal, memory, or network [terminal]?
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 100
Switch(config-vlan)#name SE
Switch(config-vlan)#exit
Switch(config)#vlan 200
Switch(config-vlan)#name IT
Switch(config-vlan)#
```

Dedicate 8 ports for each VLAN:

```
Zulqarnain_SP21379(config)#interface range fa0/3-10
Zulqarnain_SP21379(config-if-range)#switchport mode access
Zulqarnain_SP21379(config-if-range)#switchport access vlan 100
Zulqarnain_SP21379(config-if-range)#esit
                                            ^
% Invalid input detected at '^' marker.

Zulqarnain_SP21379(config-if-range)#exit
Zulqarnain_SP21379(config)#interface range fa0/11-18
Zulqarnain_SP21379(config-if-range)#switchport mode access
Zulqarnain_SP21379(config-if-range)#switchport access vlan 200
Zulqarnain_SP21379(config-if-range)#
```

Configure trunk ports between switches and the router:

```
Zulqarnain_SP21379(config)#inter
Zulqarnain_SP21379(config)#interface r
Zulqarnain_SP21379(config)#interface range fa0/1-2
Zulqarnain_SP21379(config-if-range)#sw
Zulqarnain_SP21379(config-if-range)#switchport mode access
Zulqarnain_SP21379(config-if-range)#sw
Zulqarnain_SP21379(config-if-range)#exit
Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#
```

Step 3: Network Design

Computers can communication using all VLANs (VLAN1, VLAN 100 for SE and VLAN 200 for IT) are allowed in the network.

```
Zulqarnain_SP21379(config)#inter
Zulqarnain_SP21379(config)#interface f0/1
Zulqarnain_SP21379(config-if)#sw
Zulqarnain_SP21379(config-if)#switchport trunk allowed vlan 1,100,200
Zulqarnain_SP21379(config-if)#sw
Zulqarnain_SP21379(config-if)#switchport mode access
Zulqarnain_SP21379(config-if)#
Zulqarnain_SP21379(config-if)#
```

Inter-VLAN communication must be facilitated using a single router-on-a-stick configuration.

```
Router(config)#
Router(config)#interface gigabitEthernet 0/0.100
Router(config-subif)#encapsulation dot1Q 100
Router(config-subif)#ip address 172.79.1.1 255.255.255.0
Router(config-subif)#exit
Router(config)#interface gigabitEthernet 0/0.200
Router(config-subif)#encapsulation dot1Q 200
Router(config-subif)#ip address 172.79.2.1 255.255.255.0
Router(config-subif)#exit
Router(config)#
```

Native VLAN communication should be set to VLAN 100

```
Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#interface fa0/1
Zulqarnain_SP21379(config-if)#switchport trunk native vlan 100
Zulqarnain_SP21379(config-if)#exit
Zulqarnain_SP21379(config)#
```

Step 4: IP Addressing

sub-netted Class B network (172.79.0.0/27)

```
Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#inter
Zulqarnain_SP21379(config)#interface vlan 100
Zulqarnain_SP21379(config-if)#ip address 172.79.1.1 255.255.255.224
Zulqarnain_SP21379(config-if)#
Zulqarnain_SP21379(config-if)#exit
Zulqarnain_SP21379(config)#interface vlan 200
Zulqarnain_SP21379(config-if)#ip address 172.79.2.1 255.255.255.224
Zulqarnain_SP21379(config-if)#exit
Zulqarnain_SP21379(config)#
```

Step 5: Security Constraints:

a. Console and VTY access on switches and routers must be password.

```
Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#line vty 0 4
Zulqarnain_SP21379(config-line)#password 21379
Zulqarnain_SP21379(config-line)#login
Zulqarnain_SP21379(config-line)#exit
Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#line con
Zulqarnain_SP21379(config)#line console 0
Zulqarnain_SP21379(config-line)#password 21379
Zulqarnain_SP21379(config-line)#
Zulqarnain_SP21379(config-line)#
Zulqarnain_SP21379(config-line)#exit
Zulqarnain_SP21379(config)#
```

b. Privileged EXEC mode must require a password on all network devices.

```
Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#enable se
Zulqarnain_SP21379(config)#enable secret 21379
Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#
```

c. Give each Switch and a Router a MOTD of your own choice. Min 5, Max 10 Lines. Other than message it should show your Name, Roll No. and Department Name.

```
Zulqarnain_SP21379(config)#banner motd #
Enter TEXT message.  End with the character '#'.
************************************************
Unauthorized access is strictly prohibited!
Name:  Zulqarnain Hassan
Roll No: SP21379
Department: software engineering
Violators will be prosecuted to the fullest extent!
***************************************************
#

Zulqarnain_SP21379(config)#
Zulqarnain_SP21379(config)#
```

i. Pings to all VLANS.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 172.79.1.4

Pinging 172.79.1.4 with 32 bytes of data:

Reply from 172.79.1.4: bytes=32 time=1ms TTL=128
Reply from 172.79.1.4: bytes=32 time<1ms TTL=128
Reply from 172.79.1.4: bytes=32 time<1ms TTL=128
Reply from 172.79.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 172.79.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.79.2.36

Pinging 172.79.2.36 with 32 bytes of data:

Request timed out.
Reply from 172.79.2.36: bytes=32 time=10ms TTL=127
Reply from 172.79.2.36: bytes=32 time<1ms TTL=127
Reply from 172.79.2.36: bytes=32 time=1ms TTL=127

Ping statistics for 172.79.2.36:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 172.79.2.33

Pinging 172.79.2.33 with 32 bytes of data:

Request timed out.
Reply from 172.79.2.33: bytes=32 time=28ms TTL=127
Reply from 172.79.2.33: bytes=32 time<1ms TTL=127
Reply from 172.79.2.33: bytes=32 time<1ms TTL=127

Ping statistics for 172.79.2.33:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 28ms, Average = 9ms

C:\>
```

ii. Telnet from switch (sw-2) vlan 200 pc39 to neighbor switche (sw-10 vlan 100.

```
C:\>
C:\>
C:\>
C:\>telnet 172.79.1.1
Trying 172.79.1.1 ...Open


User Access Verification

Password:
Zulqarnain_SP21379_sw1>en
Password:
Zulqarnain_SP21379_sw1#
Zulqarnain_SP21379_sw1#
Zulqarnain_SP21379_sw1#
```

iii. Pings to the designated VLAN’s Gateways.

```
C:\>
C:\>
C:\>ping 172.79.2.1

Pinging 172.79.2.1 with 32 bytes of data:

Reply from 172.79.2.1: bytes=32 time<1ms TTL=255
Reply from 172.79.2.1: bytes=32 time<1ms TTL=255
Reply from 172.79.2.1: bytes=32 time<1ms TTL=255
Reply from 172.79.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.79.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 172.79.1.1

Pinging 172.79.1.1 with 32 bytes of data:

Reply from 172.79.1.1: bytes=32 time<1ms TTL=255
Reply from 172.79.1.1: bytes=32 time<1ms TTL=255
Reply from 172.79.1.1: bytes=32 time=1ms TTL=255
Reply from 172.79.1.1: bytes=32 time<1ms TTL=255

Ping statistics for 172.79.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 172.79.1.1
```

PC configurations

i. Running configurations.

```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..: 
   Link-local IPv6 Address.........: FE80::201:43FF:FE39:9A95
   IPv6 Address....................: ::
   IPv4 Address....................: 172.79.1.18
   Subnet Mask.....................: 255.255.0.0
   Default Gateway.................: ::
                                     172.79.1.1

Bluetooth Connection:

   Connection-specific DNS Suffix..: 
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

ii. VLAN briefs.

```
Zulqarnain_SP21379_sw1#
Zulqarnain_SP21379_sw1#show vlan b
Zulqarnain_SP21379_sw1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/2, Fa0/19, Fa0/20, Fa0/21
                                                Fa0/22, Fa0/23, Fa0/24, Gig0/1
                                                Gig0/2
100  SE                               active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
200  IT                               active    Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
1002 fddi-default                     active    
1003 token-ring-default               active    
1004 fddinet-default                  active    
1005 trnet-default                    active    
Zulqarnain_SP21379_sw1#
```

iii. Trunk information.

```
Zulqarnain_SP21379_sw1#show in
Zulqarnain_SP21379_sw1#show interfaces t
Zulqarnain_SP21379_sw1#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Fa0/1       on           802.1q         trunking      100

Port        Vlans allowed on trunk
Fa0/1       1,100,200

Port        Vlans allowed and active in management domain
Fa0/1       1,100,200

Port        Vlans in spanning tree forwarding state and not pruned
Fa0/1       1,100,200

Zulqarnain_SP21379_sw1#
```